

Yongye(Felix) Hu

☎ 510-277-7666 | ✉ hyy.felix@gmail.com | 🌐 felix-hyy | 👤 felix-hu.space

EDUCATION

University of California, Berkeley

Master of Engineering | Robotics Engineering

Aug 2024 - May 2025

Berkeley, CA

Wuhan University of Science and Technology

Bachelor of Engineering (Honours) | Mechatronics Engineering

Sep 2020 - May 2024

Wuhan, China

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform

Robotics Engineer & Project Status Manager

Sep 2024 - May 2025

Berkeley, CA

- Developed a ROS2-based robotic system integrating a simulator and MPC controller for precise obstacle avoidance and full trajectory completion with real-time state logging.
- Integrated OptiTrack motion capture with the robotic platform, achieving <1 mm position accuracy and <10 ms latency for closed-loop control, enabling sub-centimeter trajectory tracking and iterative controller refinement.

ROBOCON: Multi-Robot Coordination System

Integrated Mechanical Engineer

Sep 2022 - Jul 2023

Wuhan, China

- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.
- Implemented a custom UART protocol with headers, footers, and float unpacking; leveraged DMA + interrupts for non-blocking 100 Hz updates of 6-DoF data, with <0.2% error rate.
- Modularized the robot into gimbal, power, launcher, and chassis subsystems to achieve independent functionality and decoupled operation, incorporating modular interfaces and redundancy to improve system stability and debuggability.
- Oversaw part sourcing, assembly, testing and iteration under DFA principles; delivered two autonomous launch platforms and one joystick-controlled quadruped within strict fabrication deadlines.

Lightweight, Sustainable Remote-Mining Excavator System

R&D Engineer

Jan 2021 - Jul 2023

Wuhan, China

- Designed a compact, low-power crane system with ESP32-based Wi-Fi remote control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100-MHz oscilloscope.
- Integrated a line-following module to enable autonomous navigation in semi-structured environments, reducing manual intervention by 40% and improving coordination among multiple units.

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd.

Mechatronics Engineer Intern

Sep 2023 - Mar 2024

Wuhan, China

- Reviewed and debugged Siemens S7-1200 PLC programs submitted by a third-party integrator; verified and validated the full automation workflow for surface defect detection of large flat materials.
- Integrated sensors including IR, microswitches, and pressure transducers to monitor system state and trigger alarms, enhancing operational reliability and safety.

SKILLS

Robotics & Simulation: ROS2, MPC, Closed-Loop Control, Sensor Fusion, Inverse Kinematics, PID/Fuzzy PID

Programming: Python, C/C++, Embedded C, Linux, MATLAB, Qt, LaTeX

Embedded & Hardware: STM32, ESP32, Arduino, Raspberry Pi, PCB Design (KiCad), Motor Control

Communication Protocols: CAN, UART, SPI, I2C, Modbus RTU, BLE, Wi-Fi, DMA

Tools & Prototyping: SolidWorks, Autodesk Fusion, Ansys FEA, FDM/SLA 3D Printing, CNC Machining